

DIPSAN BHATTARAI

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Master of Applied Artificial Intelligence student at Deakin University with experience in Python development, data processing, statistical evaluation, research experimentation, databases, REST APIs, and production-oriented software systems. Experienced in building reproducible experimental workflows, analyzing model performance, processing structured financial data, and developing backend systems using Python, Django REST Framework, PostgreSQL, and cloud technologies. Interested in software development, AI engineering, quantitative research and building reliable data-driven systems.

PROFESSIONAL EXPERIENCE

Research Assistant

Nov 2025 – Present

Deakin University | Burwood, VIC

- Conducting research on fine-tuning Large Language Models (LLMs) on edge devices to enable efficient on-device AI inference with reduced computational overhead.
- Investigating Federated Learning frameworks for privacy-preserving, decentralized model training across distributed environments.
- Applying optimisation strategies to enhance model performance and computational efficiency on resource constrained hardware.
- Developing experimental workflow in Python and PyTorch, including model training, evaluation, performance tracking, and result analysis.

Software Engineer Intern

Nov 2025 – Feb 2026

Deakin Applied Artificial Intelligence Institute (A2I2) | Burwood, VIC

- Designed and implemented enterprise workflow automation pipelines using Microsoft Power Platform(Power Automate, PowerBI) streamlining internal business processes in close collaboration with cross-functional stakeholders.
- Developed user-facing interfaces in Microsoft PowerApps, enabling non-technical staff to interact with and configure automated systems independently.
- Implemented AI-assisted development tools, including Claude Code, OpenAI Codex, and Vercel v0, to accelerate application prototyping and deliver responsive web solutions.
- Engineered and validated REST APIs to confirm end-to-end integration between Power Automate workflows and external third-party services.
- Collaborated with cross-functional stakeholders to gather requirements, test workflow, and deliver practical technology solutions.

AI Developer – Trainee

Jan 2025 – Jun 2025

F1Soft International Pvt. Ltd. | Lalitpur, Nepal

- Developed an intent classification system for a multinational company's customer service chatbot, employing prompt engineering and Retrieval-Augmented Generation (RAG) to enhance query-handling accuracy.
- Improved model recall through systematic performance evaluation using confusion matrices; communicated findings and proposed iterative refinements to the development team.
- Built backend services with Python and Django REST framework, and managed CI/CD deployment pipelines via Jenkins, Rancher, and ArgoCD.
- Maintained version-controlled, collaborative development using GitLab within an Agile, sprint-based team structure.

AI Intern

Nov 2024 – Jan 2025

F1Soft International Pvt. Ltd. | Lalitpur, Nepal

- Developed a Python-based multi-page bank-statement processing pipeline using PaddleOCR to extract, clean, structure, and validate financial transaction data for an internal fraud-detection project.
- Performed image annotation using Label Studio and augmentation using Python, and evaluated computer vision models using YOLO for object detection tasks.

EDUCATION

Master of Applied Artificial Intelligence (Professional)
Deakin University | Geelong, VIC, Australia
Current Grade: *Distinction*

Bachelor of Technology – Computer Science and Engineering
Vellore Institute of Technology (VIT) | Vellore, Tamil Nadu, India
GPA: 8.45 / 10

Jul 2025 – Jul 2027 (Expected)

Aug 2020 – Aug 2024

PROJECTS

- Customer Prioritisation in E-Commerce Using Generative AI**
- Engineered a novel ‘priority value’ formula incorporating order frequency and total customer spend with configurable weights, enabling dynamic customer ranking by business value.
 - Used Python and data-analysis libraries to clean transaction data, calculate customer-level metrics, rank customers and evaluate the sensitivity of rankings to different weights.
 - Deployed the solution as an interactive web application, enabling users to upload datasets and explore customer rankings, top brands, and individual spending patterns in real time.
- SecureBank – AI-Powered Banking Chatbot on GCP**
- Architected and deployed a secure, cloud-native banking chatbot on Google Cloud Platform, designed to handle customer queries with contextual accuracy.
 - Integrated Google Cloud Model Armour to detect and reject adversarial, off-topic, or potentially harmful inputs, safeguarding the system against prompt injection and misuse.
 - Delivered the complete solution including backend logic and a web-based front end within the GCP cloud environment.
- Email Tone Polisher using Microsoft PowerApps**
- Developed a Microsoft PowerApps application that rewrites emails according to configurable tones (e.g. formal, professional, assertive), with an admin interface for tone and prompt management backed by SharePoint as a data store.
 - Designed and deployed automated backend workflow pipelines in Microsoft Power Automate to orchestrate email processing logic.
 - Prototyped the user interface using Vercel v0 and Bolt.dev; constructed and validated a REST API backend via Claude Code and OpenAI Codex to confirm Power Automate connectivity with external services.

SKILLS

Languages	Python, Java, JavaScript, C/C++
Data Analysis	pandas, NumPy, Jupyter Notebook, statistical analysis, data cleaning, data visualization
AI Tools	Claude Code, OpenAI Codex, Vercel v0, Bolt.dev
Cloud & DevOps	Microsoft Azure, Google Cloud Platform (GCP), Rancher, Jenkins, ArgoCD
Databases	MySQL, PostgreSQL, Redis, Qdrant
Platforms	Microsoft Power Platform, Git/GitLab, Agile/Scrum
Soft Skills	Team Collaboration, Communication, Analytical Thinking, Adaptability, Problem Solving

CERTIFICATIONS & MEMBERSHIPS

- Microsoft Certified: AI Fundamentals (AI-901) – **Microsoft**
- Machine Learning Specialization – **DeepLearning.AI / Stanford University**
- Java Development on Oracle Cloud – **Oracle**
- Google Cloud Fundamentals: Core Infrastructure – **Google / Coursera**
- High Performance Coding Training – **Iamneo**
- Member – **Deakin Data Science Club**
- Active attendee – **Google Developer Group (GDG), Melbourne**